REVIEW

Parkinson's Disease and Alpha Synuclein: Is Parkinson's Disease a Prion-Like Disorder?

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ABSTRACT: Altered protein handling is thought to play a key role in the etiopathogenesis of Parkinson's disease (PD), as the disorder is characterized neuropathologically by the accumulation of intraneuronal protein aggregates (Lewy bodies and Lewy neurites). Attention has particularly focused on the α-synuclein protein, as it is the principal component of Lewy pathology. Moreover, point mutations in the α-synuclein gene cause rare familial forms of PD. Importantly, duplication/triplication of the wild type α-synuclein gene also cause a form of PD, indicating that increased levels of the normal α-synuclein protein is sufficient to cause the disease. Further, single nucleotide polymorphisms in the α-synuclein gene are associated with an increased risk of developing sporadic PD. Recent evidence now suggests the possibility that α-synuclein is a prion-like protein and that PD is a prion-like disease. Within cells, α-synuclein normally adopts an α-helical conformation. However, under certain circumstances, the protein can undergo a profound conformational transition to a β-sheet–rich structure that polymerizes to form toxic oligomers and amyloid plaques. Recent autopsy studies of patients with advanced PD who received transplantation of fetal nigral mesencephalic cells more than a decade earlier demonstrated that typical Lewy pathology had developed within grafted neurons. This suggests that α-synuclein in an aberrantly folded, β-sheet–rich form had migrated from affected to unaffected neurons. Laboratory studies confirm that α-synuclein can transfer from affected to unaffected nerve cells, where it appears that the misfolded protein can act as a template to promote misfolding of host α-synuclein. This leads to the formation of larger aggregates, neuronal dysfunction, and neurodegeneration. Indeed, recent reports demonstrate that a single intracerebral inoculation of misfolded α-synuclein can induce Lewy-like pathology in cells that can spread from affected to unaffected regions and can induce neurodegeneration with motor disturbances in both transgenic and normal mice. Further, inoculates derived from the brains of elderly α-synuclein–overexpressing transgenic mice have now been shown to accelerate the disease process when injected into the brains of young transgenic animals. Collectively, these findings support the hypothesis that α-synuclein is a prion-like protein that can adopt a self-propagating conformation that causes neurodegeneration. We propose that this mechanism plays an important role in the development of PD and provides novel targets for candidate neuroprotective therapies. © 2013 *Movement* Disorder Society

Key Words: misfolded α-synuclein; β-sheet formation; Lewy-like pathology; prions; toxic oligomers

There has been a surge of recent interest in the possibility that neurodegenerative diseases may be related to prion disorders.[1] The first neurodegenerative disease found to be caused by the accumulation of abnormally processed proteins was scrapie in sheep. Prion diseases in humans include Creutzfeldt-Jakob disease, Gerstmann-Sträussler-Scheinker disease, fatal insomnia, and kuru. The fundamental event in the biology of a prion is a conformational transition that converts the normal cellular protein (eg, cellular prion protein [PrP^C]) to a misfolded isoform (eg, PrP^{Sc}). These, in turn, polymerize into oligomers and amyloid fibrils that coalesce into plaques and cause neurodegeneration.[2] In this review, we provide evidence suggesting that

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α-synuclein is a prion-like protein and that Parkinson's disease (PD) is a prion-like disorder. This hypothesis suggests novel targets for the development of putative neuroprotective therapies.

Prion Disorders

Prions are infectious protein particles that lack a nucleic acid and are comprised solely of aberrantly folded proteins.[3] They cause disease by misfolding into a β-sheet–rich conformation with the formation of toxic oligomers/filaments and amyloid aggregates. The misfolded protein acts as a template to promote conformational change in the wild type protein, causing a chain reaction that ultimately leads to neurodegeneration. Further, the abnormal protein can be transmitted from affected to healthy unaffected nerve cells, in which the sequence is recapitulated, thereby extending the neurodegenerative process. In mammals, the most well-studied prion protein is PrP^{Sc}. It is formed from the precursor protein PrP^{C} by a poorly understood process in which the α-helix–rich native protein is refolded into the PrP^{Sc} protein with a high β-sheet conformation. The accumulation of PrP^{Sc} triggers the additional misfolding of host PrP^{C} (a prion conformer reaction), which join to form toxic oligomers/filaments, polymerize to form amyloid plaques, and ultimately lead to neurodegeneration with neurologic dysfunction.

Parkinson's Disease and Alpha Synuclein

That misfolded proteins might play a key role in the etiology of PD has long been considered; because, pathologically, the disease is characterized by the accumulation of aggregated proteins in the perikarya, dendrites, and axons of dopamine nerve cells in the substantia nigra pars compacta (SNc) as well as other regions of the central and peripheral autonomic nervous systems. Friederich Heinrich Lewy was the first to identify intracytoplasmic proteinaceous inclusions in the dorsal motor nucleus of the vagus and the substantia innominata of PD patients.[4] They were named Lewy bodies in his honor by Tretiakoff, who identified these inclusions in SNc neurons.[5] Similar protein aggregates were subsequently described in the axons of affected neurons and are known as Lewy neurites. Lewy bodies are comprised of a granular core that includes a wide variety of nitrated, phosphorylated, and ubiquitinated proteins surrounded by a filamentous halo primarily comprised of neurofilament and α-synuclein (Fig. 1). The precise reason why Lewy bodies develop is not known. While initially thought to be toxic and to contribute to neurodegeneration, they have more recently been postulated to represent a form of aggresome that develops in response to increased levels of misfolded proteins to segregate and facilitate the clearance of these potentially toxic proteins.[6] In support of this concept, a recent postmortem study on aged individuals with incidental Lewy body disorder provides evidence that Lewy bodies are not the first manifestation and are not a prerequisite for neurodegeneration.[7] Specifically, the investigators found Lewy bodies in neurons of the olfactory system and caudal brainstem, but did not find them in SNc neurons despite reduced tyrosine hydroxylase (TH) immunoreactivity, suggesting that the pathological process had already begun and that Lewy pathology was not the initial event.

The concept that α-synuclein is likely to be involved in the etiology of PD was first suggested following the discovery that mutations in the encoding gene were

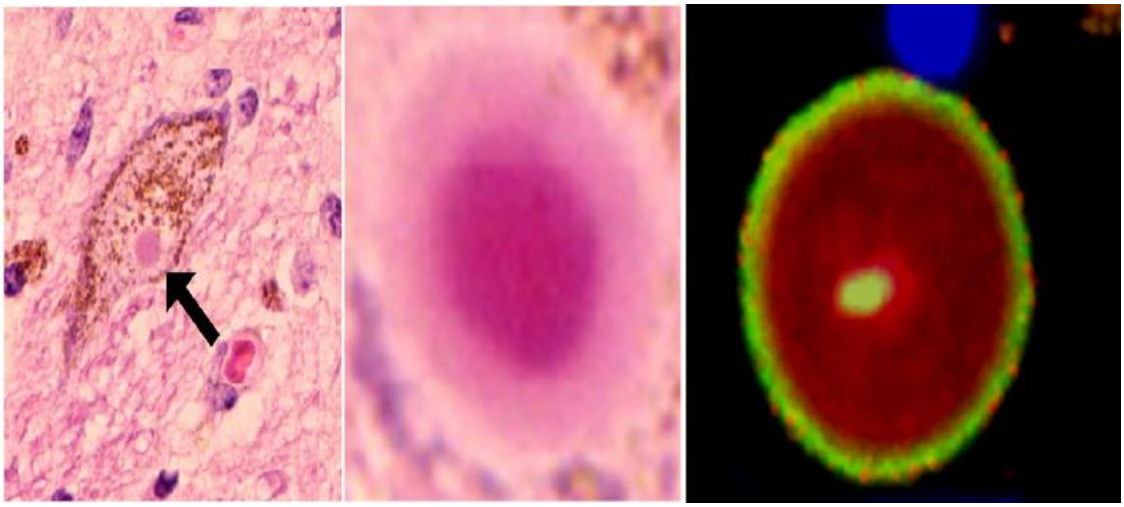

FIG. 1. (**Left**) This Lewy body in a melanized SNc dopamine neuron was stained for hematoxylin and eosin. (**Middle**) This is a high-power view of a granular core made up of various nitrated, phosphorylated, and ubiquitinated proteins surrounded by a pale halo comprised of neurofilaments and α-synuclein. (**Right**) This is a confocal image of a dual-labeled SNc Lewy body that was stained for ubiquitin (red) and α-synuclein (yellow).

associated with rare familial cases of PD.[8] Soon thereafter, it was recognized that α-synuclein was also a major component of Lewy pathology in patients with the more common sporadic form of the disease.[9] Importantly, it has also been discovered that duplication or triplication of the wild type α-synuclein gene itself can lead to a familial form of PD.[10–12] This observation suggests that overexpression of the normal, wild type α-synuclein protein itself can lead to the development of PD.

A series of clinical and laboratory observations supports the hypothesis that increased expression levels of α-synuclein are important for PD pathogenesis. More profound pathology and an earlier age of onset are noted in patients who have triplication compared with duplication of the α-synuclein gene. Cytoplasmic levels of α-synuclein increase in the aging human brain, and age is the major risk factor for the development of PD. Single nucleotide polymorphisms associated with α-synuclein are linked to an increased PD risk in genome-wide association studies.[13,14] In the laboratory, overexpression of α-synuclein in transgenic rodents can lead to loss of nigral dopaminergic neurons and the accumulation of α-synuclein aggregates.[15] Finally, overexpression of α-synuclein via gene delivery to the region of the SNc leads to the formation of inclusion bodies that stain for α-synuclein, degeneration of dopamine neurons, and parkinsonian motor disturbances in both rats and monkeys.[16]

With the introduction of stains that recognize α-synuclein, it has become evident that Lewy pathology in PD is much more widespread than just in SNc dopamine neurons. It is now appreciated that Lewy pathology affects norepinephrine neurons of the locus coeruleus, cholinergic neurons of the nucleus basalis of Meynert, serotonin neurons of the median raphe, and specific sets of nerve cells in the olfactory system, neocortex of the cerebral hemispheres, upper and lower brainstem, spinal cord, and peripheral autonomic nervous system.[17] Of particular interest with regard to the prion hypothesis are the observations of Braak and colleagues. They used α-synuclein staining to assess the distribution of Lewy pathology in a series of patients with PD and age-matched control individuals and suggested that Lewy pathology evolves in a sequential and predictable fashion, beginning in the olfactory system, peripheral autonomic nervous system, and dorsal motor nucleus of the vagus; extending to involve dopamine neurons of the SNc in the midstage of the disease; and affecting the cerebral hemispheres in the later stages of the illness.[18] While this precise sequence of degeneration may not apply to all cases,[19] it raises the possibility that neurodegeneration and Lewy pathology might extend from affected to unaffected regions in a prion-like manner.

Is Alpha Synuclein a Prion-Like Protein?

Observations in Grafted Patients With Parkinson's Disease

The possibility that α-synuclein might be a prion was dramatically highlighted by postmortem studies in PD patients who had received grafts of fetal mesencephalic brain tissue.[20,21] When PD patients died 18 months after the transplant procedure, autopsy studies demonstrated robust survival of grafted neurons that had a normal appearing dopamine neuronal phenotype, normal staining for TH and dopamine transporter, and provided a rich organotypic innervation of the denervated striatum.[22] In contrast, in patients who died 11 to 16 years after the transplant procedure, surviving grafted neurons contained pathologic Lewy body-like protein inclusions.[23,24] The inclusions stained positively for α-synuclein, ubiquitin, and a specific form of post-translationally modified α-synuclein that is phosphorylated on serine residue 129 and is normally found only in Lewy bodies and Lewy neurites (Fig. 2). In addition, the aggregates stained for thioflavin-S, a marker of a β-sheet–rich structure.[23] Thus, the inclusions in grafted neurons were identical to the typical Lewy bodies and Lewy neurites seen in PD. Furthermore, the grafted neurons demonstrated reduced staining for the dopamine transporter and, to a lesser degree, for TH with relatively normal staining for vesicular monoamine transporter 2.[23,25] These findings suggest that the grafted nerve cells were dysfunctional and that the pattern of loss of dopamine markers is similar to what is observed in PD.

Inclusions were detected in 2% to 8% of grafted neurons, approximately the same percentage detected in SNc neurons of the patients with PD. In one patient who had received grafts at two different time points (4 years apart), Lewy bodies were detected in 1.9% of pigmented neurons in the 12-year-old grafts and in 5% in the 16-year-old transplant. Levels of nonaggregated, soluble α-synuclein also were altered in grafted neurons. Normally, in TH-immunoreactive nigral dopamine neurons from individuals aged 20 years or younger, α-synuclein immunoreactivity is not detectable. During normal aging, however, an increasing proportion of nigral dopamine neurons exhibits α-synuclein immunoreactivity in the perikarya. In the grafted neurons, soluble α-synuclein was clearly detectable in many grafted neurons despite their very young age; for example TH immunoreactivity was noted in 40% of 12-year grafted neurons and in 80% of 16-year grafted neurons. Thus, the accumulation of soluble α-synuclein and Lewy body formation appeared to be time-dependent.

Because Lewy bodies and neurites were detected in embryonic nerve cells derived from multiple genetically unrelated donors, it seems clear that the

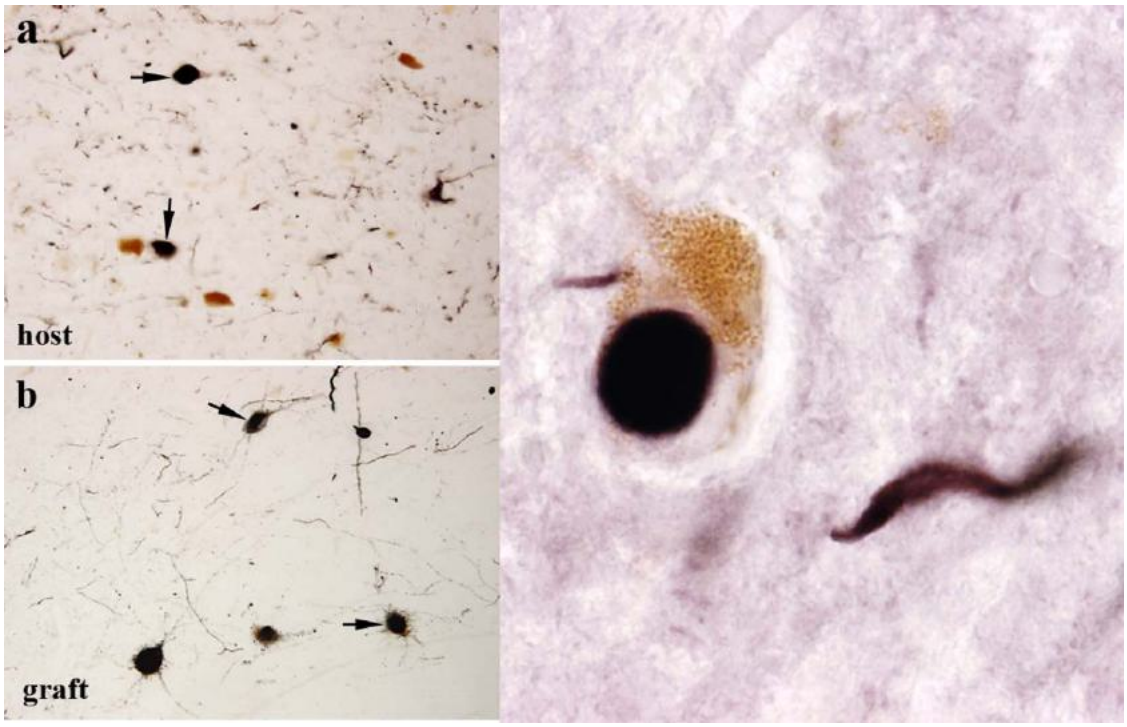

FIG. 2. (**Left**) Alpha-synuclein–stained Lewy bodies and Lewy neurites are shown in (**a**) host substantia nigra neurons and (**b**) embryonic dopamine neurons that had been implanted into the striatum of a PD patient 14 years earlier. (**Right**) This is a high-power view of an α-synuclein–stained Lewy body and a Lewy neurite in grafted mesencephalic dopamine neurons.

accumulation of Lewy pathology was directly related to the grafted neurons having been placed in a PD environment. We postulate that the most likely explanation for this is the direct transmission from PD-affected striatal dopamine terminals suggesting that α-synuclein is a prion and PD is a prion disorder.[26,27] We have considered elsewhere the possibility that other processes (eg, poor trophic factor support, neuroinflammation, oxidative stress, excitotoxicity) also might have played a contributory role,[23,28] but we consider the prion hypothesis the most likely explanation, as described below.

Cell and Animal Models Provide Support for the Prion-Like Hypothesis

Laboratory studies provide strong support for the prion hypothesis. In vitro studies demonstrate that α-synuclein monomers and aggregates can be secreted from affected neurons via exocytosis and taken up by unaffected neurons via endocytosis.[29,30] Desplats et al. further demonstrated that transgenic mice expressing mutant human α-synuclein can transfer α-synuclein from host cells to mouse cortical neuronal stem cells that had been implanted into the hippocampus or striatum.[31] Hansen et al. similarly demonstrated the transfer of α-synuclein between host cells and grafted cells in mice that overexpressed human α-synuclein.[32] However, in contrast to the study by Desplats et al., Hansen et al. observed this phenomenon in postmitotic embryonic dopaminergic neurons that had been grafted into the striatum. Thus, their findings more closely replicate the observations in grafted patients with PD. Similar results were observed in rats that overexpressed human α-synuclein after adenoviral gene transfer.[33]

The previous experiments demonstrated α-synuclein transfer in transgenic animals that overexpressed human α-synuclein. Luk et al. demonstrated that exposure to preformed fibrils generated from purified, recombinant, human wild type α-synuclein also induces the expression of aggregates and Lewy body-like pathology in cultured cells that overexpress wild type α-synuclein.[34] They demonstrated that these α-synuclein fibrils "seeded" recruitment of endogenous α-synuclein protein and promoted its conversion into insoluble, hyperphosphorylated, and ubiquitinated pathological species, recapitulating the key features of Lewy bodies in PD. Subsequently, this same group showed that exogenous administration of preformed α-synuclein fibrils generated from purified, recombinant, human wild type α-synuclein could be taken up by cultured hippocampal neurons, transported to the perikarya, and induce the formation of Lewy-like inclusion bodies and neurodegeneration.[35] Importantly, in these in vitro studies, exposure to α-synuclein induced neurodegeneration in wild type neurons without the need for overexpression of either wild

type or mutant forms of the protein. Freundt and colleagues also grew cultured mouse cortical neurons in a microfluid device that allowed for separation of nerve cells and showed that fluorescent-labeled fibrillar α-synuclein added to the culture media could be taken up by axons and transported to second-order neurons.[36] As a group, these studies indicate that exposure to preformed fibrils derived from wild type α-synuclein can induce endogenous α-synuclein to misfold, form inclusions, and cause cell death in otherwise normal neurons.

A critical series of experiments further supports the notion that α-synuclein is a prion-like protein. Mougenot and colleagues inoculated the brains of young transgenic mice with brain homogenates that contained insoluble α-synuclein aggregates (immunoreactive for phosphorylated serine in residue 129) derived from older and clinically affected transgenic mice that overexpressed human A53T mutant α-synuclein.[37] These intracerebral inoculations triggered an earlier onset of behavioral abnormalities and α-synuclein aggregates in young transgenic mice compared with noninoculated mice or mice that received inoculations from young, clinically unaffected transgenic mice. Importantly, α-synuclein knockout mice that were inoculated with a brain homogenate from a sick TgM83 mouse were not affected, again indicating the importance of templating and the role of host wild type α-synuclein in this prion-like process. Indeed, the authors concluded that their findings were consistent with "prion-like" propagation of the disease.

Luk and coworkers took these findings one step further.[38] They demonstrated that Lewy-like α-synuclein aggregates also formed after a single injection of preformed α-synuclein fibrils into the striatum or cortex of wild type mice. Importantly, in contrast to the earlier studies performed in transgenic A53T α-synuclein mice, these mice did not overexpress mutant or wild type α-synuclein, indicating that these prion-like events could occur in normal animals if they were exposed to the presumed prion-like α-synuclein species.[38] Moreover, the study showed that, after intrastriatal injections of the preformed α-synuclein fibrils, a subgroup of nigrostriatal dopamine neurons had α-synuclein inclusions and no longer stained for TH, consistent with retrograde transport of toxic α-synuclein fibrils to the nigra from the injection site. In these animals, striatal dopamine levels were reduced and the mice developed motor deficits. There were no signs of aggregate formation or nigrostriatal dysfunction when similar preformed recombinant α-synuclein fibrils were injected into α-synuclein null mutant mice, again indicating that α-synuclein from the host brain was essential for the pathology to develop, supporting the prion hypothesis. Notably, in that study, all of the Lewy pathology developed in brain regions anatomically connected to the site of injection, ie, the striatum. Regions that were not connected to the striatum (eg, septum, hippocampus, and cerebellum) were devoid of α-synuclein aggregates.

Several studies have addressed the mechanism underlying intercellular transfer of α-synuclein. Uptake of recombinant α-synuclein injected into the cortex of rats was partially blocked by pharmacological inhibition of endocytosis, suggesting that this mechanism was involved in the transfer of the protein between cells.[32] In a later study, the same group used an intrastriatal grafting paradigm in rats involving viral vector-mediated transfer of human α-synuclein.[39] They showed that, in cells where human α-synuclein was taken up by a transplanted neuron, it sometimes was present near the cell surface and colocalized with the early endosome antigen 1, suggesting that it had recently undergone endocytosis. The study provided the first direct evidence in vivo that imported α-synuclein can seed the aggregation of endogenous α-synuclein. Within grafted rat neurons, small aggregates that were immunoreactive to an antibody recognizing human α-synuclein were surrounded by a larger area of aggregate formation that stained with an antibody recognizing rat α-synuclein. These findings suggest that transferred α-synuclein seeds host α-synuclein to misfold and aggregate.

Another series of studies has explored whether exosomes play a role in the transfer of α-synuclein between cells. Exosomes are small vesicles 30 to 90 nm in diameter that are released by cells when multivesicular bodies fuse with the outer membrane. They can transport proteins (over 10,000 different proteins are found in exosomes), mRNA or miRNA, over large distances, thereby potentially playing an important role in cellular communication. Recent studies have suggested that exosomes containing α-synuclein are released after calcium stimulation and that they can be internalized by, and become toxic to, cultured neurons.[40] The α-synuclein that is associated with exosomes is typically oligomeric (thought to be the toxic species) and can be found both inside and on the surface of exosomal vesicles.[41] The release of exosomes from neurons to the extracellular space can be increased following inhibition of autophagy and the lysosomal system, suggesting that cell stress potentially leads to more cell-to-cell transmission of α-synuclein.[42]

Taken together the studies described above demonstrate that α-synuclein fibrils/aggregates can be taken up by neuronal axons, undergo axonal transmission, promote the misfolding and fibrillization of host α-synuclein, induce the formation of inclusions in axons and cytoplasm of SNc dopaminergic neurons, lead to a reduction in striatal dopamine and neuronal dysfunction, cause neurodegeneration with motor

abnormalities, and, under certain circumstances, be transmitted to unaffected healthy neurons to extend the neurodegenerative process. Further, intracerebral inoculates of misfolded α-synuclein can induce or accelerate neurodegeneration in both transgenic and wild type mice. These findings are consistent with the possibility that α-synuclein is a prion-like protein and support the possibility that PD is a prion-like disorder. They provide a possible explanation for why embryonic dopamine neurons implanted into the striatum of PD patients develop Lewy body pathology, and they may account for the sequential pathologic changes that appear to occur in at least some PD cases. The precise α-synuclein species that is toxic and causes neurodegeneration is not known, but oligomers have been implicated. Recent studies by Cremades et al. used fluorophores to divide α-synuclein into two distinct types of oligomers; type A and B.[43] The B oligomers were proteinase K-resistant, implying β-sheet formation, and were thought to be the toxic species. Those authors proposed that misfolded α-synuclein species initially join to form loosely bound type A intermediates, which are then converted to type B oligomers in what they consider to be the critical step in inducing toxicity.

Similarities between Alpha Synuclein and Cellular Prion Protein

With the wealth of data emerging from in vitro and in vivo studies, it is evident that the events associated with PrP^{Sc} formation and neurodegeneration are mirrored by α-synuclein and PD (Table 1). Both PrP and α-synuclein natively exist in an α-helix–rich conformation state[44] but can misfold to form a β-sheet–rich protein, particularly when present in high concentration or in mutant form. In both cases, the native protein resists aggregation, but the misfolded protein is prone to assemble into fibrils, aggregates, and amyloid plaques that are associated with neurodegeneration. Further, it now appears that, like PrP, α-synuclein can be transferred from affected cells to healthy cells, where it can propagate the neurodegenerative process. As with PrP, the normal function of α-synuclein is not precisely known. α-Synuclein is primarily found in the region of the synapse and is thought to play a role in vesicular transport and facilitation of dopamine release[45]; however, like PrP, knock out of α-synuclein is not associated with major behavioral alterations in mice.[46]

What Initiates Misfolding of Alpha Synuclein?

An important question remains about what induces α-synuclein to misfold in PD. Mutant forms of the protein are prone to misfold, but mutations in the α-synuclein gene account for only a very small number of familial cases and do not account for the majority of cases that occur sporadically. Duplication or triplication of the wild type gene is also associated with familial PD. While such cases are extremely rare, they indicate that increased levels of the normal protein itself can cause the disease. In sporadic cases of PD, increased levels of α-synuclein might occur as a consequence of impaired clearance of the protein. α-Synuclein is normally cleared by the autophagy/lysosomal and ubiquitin proteasome systems,[47–51] and defects in both of these systems have been detected in patients with sporadic PD.[52,53] Moreover, inhibition of proteasomal or lysosomal clearance in experimental models is associated with neurodegeneration and the formation of inclusion bodies that stain for α-synuclein[54] as well as an increased likelihood that α-synuclein is secreted in exosomes.[42] In addition, mutations in genes that encode for proteins involved in lysosomal function (glucosidase β acid [GBA]; scavenger receptor class B, member 2 [SCARB2]; ATP13A2) and proteasomal function (Parkin, ubiquitin carboxy-terminal hydrolase L1 [UCH-L1]) are associated with, or are risk factors for, the development of PD. These observations support the notion that impaired protein clearance could cause α-synuclein to accumulate and contribute to the development of PD and can lead to α-synuclein accumulation.[55] Furthermore, there is evidence that α-synuclein aggregates can interfere with the function of the ubiquitin proteasome and lysosomal systems,[55–58] thereby further impairing clearance of the protein. Thus, one could envision a vicious cycle whereby damage to proteasomal and lysosomal systems could cause α-synuclein to accumulate, and increased levels of α-synuclein could inhibit the proteasome and lysosomal systems, ultimately leading to the formation of oligomers and aggregates with resultant neurodegeneration and clinical dysfunction.[59] Indeed, α-synuclein aggregate formation and alterations in markers of proteasome and lysosomal function develop simultaneously following gene delivery of α-synuclein,[60] consistent with this hypothesis.

Alternatively, increased levels of α-synuclein might arise as a consequence of damage or toxicity due to factors such as oxidative stress or inflammation that cause the protein to misfold and resist clearance. The observations by Braak et al. suggest that the earliest central nervous system changes in PD are found in olfactory structures and in the dorsal motor nucleus of the vagus nerve.[18] In this regard, it is noteworthy that the olfactory nerve terminals are in direct contact with the external environment, and terminals from the dorsal motor nucleus of the vagus nerve reside within the gastric submucosa a mere few millimeters from the lumen. Thus, these nerve terminals may be at particular risk of being exposed to infectious or toxic agents, which could cause damage and trigger an aggregation of α-synuclein protein that could spread to affect SNc dopamine neurons and other PD-affected regions in a

TABLE 1. Similarities Between Prion Diseases and Parkinson's Disease

- Both prion diseases and PD are age-dependent neurodegenerative disorders.
- Both prion diseases and PD illustrate selective vulnerability affecting specific subgroups of neurons. A misfolded isoform of cellular protein prions (PrP^{Sc}) can be confined to small regions of the brain (such as the cerebellum in the ataxic forms of CJD or GSS or the dorsal medial nuclei of the thalamus in FFI) or can be more widespread, as in most cases of sporadic CJD. In PD, α-synuclein deposits are found in selected regions (e.g. SNc, LC, DMV, etc) while sparing neighboring nuclei (e.g cerebellum, hypoglossal nuclei), or can be widespread as in cases of Lewy body dementia.
- Both diseases are ~90% sporadic and ~10% familial.
  - Point mutations in the α-synuclein and PrP genes cause inherited forms of PD and prion diseases.
  - Familial prion diseases can be caused by expanded octarepeat inserts in the PrP genes, and familial PD can be caused by duplication and triplication of the wild-type α-synuclein gene.
- Protein deposits are characteristic features of both disorders.
- Proteins in both diseases undergo an α → β conformational transition:
  - α-synuclein in PD and PrP in prion diseases.
- Proteins in the disease state acquire a high β-sheet conformation and readily polymerize into fibrils, Lewy bodies in PD, and amyloid plaques in prion diseases.
- Misfolded proteins can be taken up by neurons and transmitted to unaffected neurons.
- Misfolded proteins are thought to migrate and spread across the neuraxis in both diseases: PrP^{Sc} prions migrate along peripheral nerves and up the spinal cord, whereas it is postulated that α-synuclein migrates from the gut and olfactory regions to affect specific regions of the CNS.
- Both diseases can be transmitted in humans—from corneal grafts and dura mater to human recipients (prion diseases) and from host to implanted dopamine neurons (PD).

Abbreviations: CJD, Creutzfeldt-Jakob disease; GSS, Gerstmann-Sträussler-Scheinker disease; FFI, fatal familial insomnia; SNc, substantia nigra pars compacta; LC, locus coeruleus; DMV, dorsal motor nucleus of the vagus.

prion-like manner.[27] Indeed, constipation is known to be over-represented among individuals who later develop PD,[61] and colonic biopsies obtained years prior to the onset of PD features demonstrate α-synuclein aggregates in colonic submucosal neurons.[62] Work in experimental animals has highlighted the possibility that the gut could be a location where α-synuclein aggregation first occurs in PD. Pan-Montojo et al. administered the mitochondrial toxin rotenone locally into the intestine and reported α-synuclein aggregation in the intestinal wall that, over time, propagated to and caused neurodegeneration in the dorsal motor nucleus of the vagal nerve and eventually in the substantia nigra.[63] In a follow-up study, the same team reported that cutting the vagal nerve or partially removing sympathetic ganglia could block the spreading of Lewy-like pathology from the gut to the central nervous system.[64] Lee and coworkers have also addressed the idea that the gut could be a starting point for α-synuclein misfolding. They showed that injection of brain extracts prepared from patients with dementia with Lewy bodies, but not from normal brains, induced the deposition of α-synuclein aggregates in myenteric neurons of transgenic mice that overexpressed human A53T α-synuclein.[65] Although all of those studies need to be replicated, they should stimulate further investigations into whether the intestine (or olfactory system) could represent an initial anatomical site where α-synuclein misfolds and triggers the PD process.

It is also possible that α-synuclein might stochastically undergo a conformational transition from an α-helix–rich structure to one high in β-sheet structure at a low frequency under normal circumstances. If such a stochastic process were to escape physiological control by the protein clearance systems, this could result in increased levels of misfolded protein, which could then act as a template to promote misfolding of native wild type α-synuclein and a prion-like chain reaction leading to neurodegeneration. The finding that duplication or triplication of the wild type α-synuclein gene can lead to PD suggests that a 50% increase in α-synuclein levels is sufficient to drive the protein into an alternative conformational state capable of inducing the disease. The proposed stochastic nature of the process in which α-synuclein adopts a β-sheet–rich conformation at a low frequency could explain the age-dependent nature of sporadic PD, as there is an age-related reduction in proteasomal/lysosomal function[66] and an increase in α-synuclein protein accumulation.[67] This concept is similar to what is found in sporadic prion diseases, in which the formation of PrP^{Sc} appears to be a stochastic process, albeit one that appears to occur much less frequently than PD.

Thus, although genetic mutations represent an obvious source of increased levels of aberrantly folded α-synuclein in patients with familial PD, a combination of stochastic, aging, inflammation, environmental toxins, and genetic risk factors all could contribute to the increased production and/or impaired clearance of α-synuclein, leading to misfolding of the protein and, ultimately, neurodegeneration and the clinical features of PD.

Conclusions and Implications for a Neuroprotective Therapy

A growing body of evidence supports the hypothesis that α-synuclein is a prion and that PD is a prion

disorder. This hypothesis suggests that native α-synuclein undergoes a conformational change, promotes misfolding of the wild type protein in a chain reaction, forms toxic oligomers, and polymerizes to form amyloid plaques. This sequence eventually causes neuronal dysfunction and neurodegeneration. Furthermore, there is evidence that aggregated α-synuclein from affected neurons can be transferred to unaffected neurons to extend the neurodegenerative process, ultimately leading to the clinical features of PD.

If α-synuclein can be confirmed to be a prion-like protein and PD can be confirmed as a prion-like disorder, then it will revolutionize our thinking about the etiology of PD and provide novel targets for candidate neuroprotective therapies.[55] These could include therapies that maintain α-synuclein in a stable tetrameric form that resists misfolding, which, despite some controversy,[68] has been suggested to be its native form.[44] Another approach could be to knock out native α-synuclein and, thus, eliminate the protein substrate for misfolding and β-sheet formation. Indeed, in experimental models, α-synuclein transfer does not induce aggregate formation or neurodegeneration in α-synuclein null cells or animals (see above). As α-synuclein has not yet been shown to have a critical biological function when knocked out during embryonic development, this approach might be feasible, although it remains to be established that eliminating α-synuclein from adult neurons is not toxic.

Immunization therapy with human α-synuclein has been shown to reduce α-synuclein aggregate formation and reduce neurodegeneration in human α-synuclein transgenic mice.[69] A subsequent series of cell culture and animal experiments suggests that antibodies against α-synuclein reduce cell-to-cell transfer of the protein by directing extracellular α-synuclein to microglia, where it can be degraded.[70] Other approaches might include therapies that interfere with molecules involved in the prion conformer reaction, whereby the misfolded protein promotes the conversion of the wild type α-synuclein protein to form a β-sheet configuration, gene therapies that enhance the function of the lysosomal or proteasomal systems, and agents that promote the conversion of oligomers from a toxic to a nontoxic form.

Currently, there is no evidence for transmission of PD from one individual to another. However, there may be a very long latency (11–16 years in the case of embryonic cells transplanted into PD patients), and PD is a relatively common disease, affecting 1% to 2% of individuals over age 60 years. Thus, it may be very difficult to conduct such a study. Until it is confirmed whether or not α-synuclein is a prion and whether PD is a prion disorder, it might be wise to perform studies with human brain tissue and inoculation studies in transgenic animals in biosafety level 2-rated animal facilities.

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